

2.5 hours

Problem 1 (EVANSASY)

We claim that the answer is the shoemaker's knife.

We consider the image of the inversion of the circle centered at $(6, 2k)$ of radius 1 about the circle centered at $(0, 0)$ of radius $\sqrt{35}$. Let X, Y be the points of tangency from A to the circle. Note that,

$$AX = AY = \sqrt{d^2 - 1}$$

Using similar triangles, the midpoint of XY is at a distance $\frac{d^2-1}{d}$ from point A . Under inversion, X^*, Y^* satisfy $AX^* = AY^* = \frac{35}{\sqrt{d^2-1}}$. The midpoint of XY inverts to the center of the new circle at a distance of $\frac{35d}{d^2-1}$ from the midpoint. We may check that this is the circle described.

Inverting back, by the Shoemaker's knife, this would be the k -th circle tangent to the semicircles and the previous circles.

Problem 3 (11BAMO4)

We invert at D . A_1, B_1, C_1 would then be where cevians A^*D, B^*D, C^*D intersect the opposite side.

$$\begin{aligned} \frac{AD}{AA_1} + \frac{BD}{BB_1} + \frac{CD}{CC_1} &= 1 \\ \Leftrightarrow \sum \frac{\frac{r^2}{A^*D}}{A^*A_1 \cdot \frac{r^2}{A^*D \cdot DA_1^*}} &= 1 \\ \Leftrightarrow \sum \frac{DA_1^*}{A^*A_1^*} &= 1 \end{aligned}$$

This is clearly true.

5 clubs; 3.5 hours

Problem 5 (13SRB3)

We invert about a circle with radius $\sqrt{\frac{bc}{2}}$ centered at A followed by a reflection over the interior angle bisector of $\angle A$. Note that B, P and C, N respectively trade places. Since line AO inverted then reflected would become the altitude from A to BC , and,

$$90 = \angle APO = \angle AO^*P^* = \angle AO^*B$$

We have O goes to is the base of the altitude from A to BC . AM will be inverted then reflected to the symmedian and M will lie on (APN) . We may check that (BM^*OC) is cyclic.

In conclusion, (ABO) inverts to the nine-point circle of ABC and (MNP) the nine-point circle inverts to (ABO) . Therefore, X, Y swap places meaning $\angle BAX = \angle CAY$ must be true.

Problem 6 (18INMO3)

We invert at A . Circles Γ_1, Γ_2 get sent to lines that intersect at B^* while O_1^*, O_2^* get inverted to the reflection of A over those lines. Therefore, B^* is the circumcenter of triangle $AO_1^*O_2^*$.

Claim: $(AB^*D^*O_1^*)$ is cyclic.

Proof: $\angle AO_1^*O_2^* = \frac{\angle AB^*O_2^*}{2} = \angle AB^*D^*.$ $\square$

We have B, D, O_1 is collinear and similarly would C, B, O_2 be. We know E, F are the reflections of B over O_1, O_2 so by power of a point we are done.

9 clubs; 4.5 hours

Problem 7 (95RUSEL)

Let $P = AC \cap BD$. We invert about the semicircle. A, B, C, D stay fixed, K goes to P , while M goes to the intersection of (AOD) with AB .

We have $\angle OKM = \angle K^*M^*O$ so we are done by showing (COD) intersects AB again at the base of the altitude from P to AB . This is clearly true since (COD) is the nine-point circle of APB .

Problem 9 (18TMD8)

Let $P = AC \cap BD$. Let P^* be the image of P under inversion by $(ABCD)$. Let the tangent to $(ABCD)$ at A meet BD at Y^* .

Claim: AY^*P^*P is cyclic.

Proof: Note that P^* is the second intersection of OP with (AOC) . By similar triangles,

$$\angle AP^*P = \angle ACO = \angle OAC = \angle AYP \quad \square$$

We now invert the entire diagram under $(ABCD)$. We know A, B, C, D remain fixed, P^* and P swap places, lines AY, CZ go to the circles passing through O tangent to $(ABCD)$ at A, C respectively. The two circles intersect each other at the midpoint X of AC and let them intersect (BOC) at Y, Z respectively. We know (XYZ) is the image of the circumcircle of Δ . In particular, Y, Y^* invert to each other. Lastly, (BOC) goes to BC .

We show that (XYZ) is now tangent to BD . We claim the point of tangency is P . Since $XP \perp BD$, it suffices to prove $\angle XYP = \angle XZP = 90$.

We have $\angle XYP = \angle XYO + \angle OYP = \angle XAO + \angle Y^*P^*O = \angle XAO + \angle Y^*AP = 90$ as desired (using cyclic claim along the way). The conclusion for $\angle XZP = 90$ follows analogously and we are done.

16 clubs; 6 hours

Problem 10 (14NIMOW7)

We invert about the incircle. A^*, B^*, C^* go to the midpoints of EF, FD, DE respectively. $(A^*B^*C^*)$ is now the nine-point circle of DEF .

Note that $DP \perp EF \Leftrightarrow P \in (A^*B^*C^*) \Leftrightarrow P^* \in (ABC)$. Note that P^* is where line IQ intersects (EIF) again. It suffices to prove that if P' is the intersection of $(AEIF)$ and (ABC) other than A , we have P', I, Q are collinear. Indeed, since $\angle ABQ = \angle ACQ = 90$ implies $(ABQC)$ is cyclic with diameter AQ ,

$$\angle AP'I = \angle AP'Q = 90$$

Problem 12 (18ELMO3)

We invert at A . We now have a circle passing through A and can draw any circle or line not passing through A . We retain the ability to mark arbitrary point on lines or circles and intersections.

We draw arbitrary points B, C, D, E on the circle. We draw the complete quadrilateral and let $BD \cap CE = P; BC \cap DE = Q; CD \cap BE = R$. We may additionally have $(QBE), (QCD)$ intersect at P^* . Since $(BECD)$ is cyclic, we have P^* its Miquel Point is the inversion of P about $(BCDE)$.

Doing the same on quadrilaterals $BDCE$, we may construct the inversion of Q over $(ABCDE)$. Intersecting PP^*, QQ^* , we'd get O the center of $(ABCDE)$. The inversion of O over the circle at A , is the reflection of A over l which is one of our desired points.

For the second part, note that since we may never do anything with A , we may treat A the same as any other arbitrary point on the circle. This means we may never construct the A -antipole like how we may never construct a fixed point that we arbitrarily chose on the circle.

24 clubs; 8.5 hours

Problem 13 (Z745C83C)

We invert at A with radius $\sqrt{bc}$ and then reflect across the angle bisector. This transformation swaps $B, C; \overline{BC}, (ABC)$. The A -mixtilinear circle hence gets sent to the A -excircle.

Claim 1: I goes to the excentre.

Proof: It is a well-known fact that $AI \cdot AI_A = (\sqrt{bc})^2 = bc$. The reflection did not affect I as it was already on the angle bisector.

Using the claim, we see that, before the inversion, I was the midpoint of KL since K, L are the points of tangencies. This completes part A . For part B , we show that AT goes to the A -Nagel cevian after the transformation. This is clearly true since T went to the point of tangency E of the A -excircle at BC . For part C , $L^* = AL \cap BC$. We may check that (L^*AEI_A) is cyclic. Since $\angle L^*AI_A = 90 = \angle L^*DI_A$.

Problem 14 (17USATST2)

We invert about the (ABC) .

Note that since $\angle TAO = 90$, the circle with diameter AT is orthogonal to (ABC) meaning it remains fixed under the inversion. We have (BOC) goes to $\overline{BC}$. Therefore, A_1, A_2 go to the intersections of the circle with diameter AT with BC .

We have $OA_1 < OA_2 \Rightarrow OA_1^* > OA_2^*$. Therefore, A_1^* is T and A_2^* is the point such that $\angle AA_2T = 90$ meaning it is the base of the altitude from A to BC . We have,

$$\begin{aligned} & AA_1, BB_1, CC_1 \text{ concurrent} \\ \Leftrightarrow & (AOA_1^*), (BOB_1^*), (COC_1^*) \text{ concurrent at point other than } O \\ \Leftrightarrow & (AOA_1^*), (BOB_1^*), (COC_1^*) \text{ have collinear centers} \end{aligned}$$

Since $\angle TAO = 90$,

$$\Leftrightarrow \text{Midpoints of } OT \text{ and analogous points for other vertices collinear}$$

Doing a homothety at O of scale two, we finish showing that the three intersections T, U, V of tangents to the circle with the opposite sides are collinear. This may be done with Menelaus', note that,

$$\begin{aligned} \frac{BT}{TC} &= \frac{TB^2}{TB \cdot TC} = \frac{TB^2}{TA^2} = \left(\frac{\sin(\angle DAB)}{\sin(\angle DBA)} \right)^2 = \left(\frac{\sin(C)}{\sin(B)} \right)^2 = \frac{c^2}{b^2} \\ \Leftrightarrow \frac{TB}{TC} &= -\frac{c^2}{b^2} \end{aligned}$$

Doing this for U, V, W , all the a, b, c cancel and their product is clearly -1 . This completes part A.

We show that $(AOA_2^*), (BOB_2^*), (COC_2^*)$ intersect on a point on Euler line OH (which has inverted to itself). Once again, A_2^*, B_2^*, C_2^* are the bases of the altitudes. Clearly, O is on the radical axes of the pairs of the circles. We have H is as well since $HA_2^* \cdot HA = HB_2^* \cdot HB = HC_2^* \cdot HC$ meaning OH is their shared radical axis. The circles hence concur again on OH the Euler line as desired.

32 clubs; 11 hours

Problem 15 (95ISR)

We invert at A with radius $\sqrt{AP \cdot AQ}$ and then reflect over the angle bisector. P, Q switch places, line AB goes to line AO the diameter through A . We have γ is now tangent to AO, P^*Q^* , and (AP^*Q^*) , touching P^*O and not Q^*O . Note that C is now where the circle touches (AP^*Q^*) . Since AO and P^*Q^* are both diameters, by symmetry, γ touches (AP^*Q^*) at C^* the arc midpoint. As a result, $\angle P^*AC^* = \angle C^*AB^*$ meaning $\angle PAC = \angle CAB$ as desired.

Problem 16 (09RUS102)

Invert at A with radius $\sqrt{bc}$ then reflect over AE the angle bisector. D, E switch places; B, C switch places. Since (DE) was already centered on the angle bisector, it is still centered on the angle bisector. Hence, the circle is fixed after the transformation. The intersection of (DE) with Ω goes to its intersection of it with BC . We have,

$$\angle AF^*D^* = 90$$

$D^* = E$'s foot onto BC is M the midpoint. Before the transformation, AF^* the median was the symmedian.

38 clubs; 12.5 hours

Problem 17 (16USATST2)

Let E be where the excircle touches BC . We invert at A with radius ..., reflect.... B, C are swapped, K, M are swapped. Therefore, $AK \cdot AM = bc$ meaning (DKM) is orthogonal to the circle we invert about.

Claim: E goes to T after the transformation.

Proof: Let TD meet (ABC) again at P . We have,

$$\angle MTP = \angle MTD = \angle AKB = \angle KAB + \angle ABK = \angle ACM$$

Therefore, P is the reflection of A over BC 's perpendicular bisector. If we reflect D, T across the perpendicular bisector, we get A, E, T' are collinear meaning $\angle CAE = \angle TAB$.

On the other hand, after the inversion, the excircle goes to the mixtilinear incircle meaning $\angle BAE^* = \angle CAE$ implies E goes to T after the transformation as desired. $\square$

Let S'_1 be where AT meets $(TDKM)$ again. After the inversion, T inverts to S'_1 . However, after the reflection, S'_1 went to E . Therefore, S'_1 is the reflection of E over AM , a line that passes through the A -excentre. As a result, S'_1 lies on the A -excircle.

Problem 19 (16CAN5)

Claim 1: DE, MQ intersect on (ABC) .

Proof: We invert about the circle with diameter AB . Note that M is the center of inversion. A, B, E, D which lie on the circle are fixed. Note that,

$$\angle AH^*B = \angle H^*AM + \angle MBH^* = \angle MHA + \angle BHM = \angle BHA = \angle ACB$$

Therefore, H inverts to a point on (ABC) . Note that P, Q , lying on $(MPEDQ)$ and (AHB) , now lie on the intersection of line DE and circle (ABC) . P^*, Q^* also lie on MP, MQ respectively so $MQ, DE, (ABC)$ concur. $\square$

Claim 2: PH, DE intersect on (ABC) .

Proof: We do an inversion at H of radius $\sqrt{\frac{1}{2}|Pow_{(ABC)}(H)|}$ followed by a reflection across H . Note that, by reflecting the orthocenter, M, E, D get sent to where the ray through itself on (ABC) . This suggests (MED) the nine-point circle goes to (ABC) while $\overline{DE}$ goes to (HAB) . We have $PH \cap DE$ is now where (ABC) and PH intersect again. Therefore, $PH \cap DE$ gets sent to point P . However, since $P \in (MED)$, P gets sent to a point on (ABC) after the transformation. $PH \cap DE \in (ABC)$ as desired.

48 clubs; 16 hours

Problem 20 (18KAZ116)

We invert at M with radius 1. We have,

$$\begin{aligned}\angle AMB &= \angle ADM + \angle BCM \\ \Leftrightarrow \angle A^*MB^* &= \angle D^*A^*M + \angle C^*B^*M\end{aligned}$$

Angle chasing,

$$\Rightarrow A^*D^* \parallel B^*C^*$$

Similarly,

$$\angle AMD = \angle ABM + \angle DCM \Rightarrow A^*B^* \parallel C^*D^*$$

We have $A^*B^*C^*D^*$ is a parallelogram.

Note that,

$$\begin{aligned}AM \cdot CM + BM \cdot DM &\geq \sqrt{AB \cdot BC \cdot CD \cdot DA} \\ \Leftrightarrow \frac{1}{A^*M \cdot C^*M} + \frac{1}{B^*M \cdot D^*M} &\geq \sqrt{\prod \frac{A^*B^*}{A^*M \cdot B^*M}} \\ \Leftrightarrow \frac{1}{A^*M \cdot C^*M} + \frac{1}{B^*M \cdot D^*M} &\geq \frac{\sqrt{A^*B^* \cdot B^*C^* \cdot C^*D^* \cdot D^*A^*}}{A^*M \cdot B^*M \cdot C^*M \cdot D^*M}\end{aligned}$$

We have $A^*B^* = C^*D^*$; $B^*C^* = D^*A^*$.

$$\begin{aligned}\Leftrightarrow \frac{1}{A^*M \cdot C^*M} + \frac{1}{B^*M \cdot D^*M} &\geq \frac{A^*B^* \cdot B^*C^*}{A^*M \cdot B^*M \cdot C^*M \cdot D^*M} \\ \Leftrightarrow B^*M \cdot D^*M + A^*M \cdot C^*M &\geq A^*B^* \cdot B^*C^*\end{aligned}$$

We now translate M by vector $\overrightarrow{AD} = \overrightarrow{BC}$ to M' . By Ptolemy we are done.

Problem 22 (15IMO3)

Let D, E be the bases of the altitudes from C, B . Let the line through M parallel to AQ intersect (DEF) again at G . Let the midpoints of AH, QH be M_A, M_Q .

We invert at H with radius $\sqrt{\frac{1}{2}|Pow_{(ABC)}(H)|}$ then reflect across H . Note that D, E, F swap with C, B, A respectively. As a result, $(AQDHE)$ swaps with BC ; (ABC) swaps with (DEF) the nine-point circle. As a result, Q swaps with M . (HKQ) , which has diameter HQ , hence goes to the line through M perpendicular to HM and K^* goes to G where this line intersects the nine-point circle.

We show that (AQG) is tangent to MG . Since $AQ \parallel MG$, it suffices to show that the line through G and the midpoint of AQ is perpendicular to MG . Since $AQ \perp QM \perp MG$, we show that $AQ = 2MG$ to finish. Note that, indeed, since $M_A M_1 \perp M_Q M \perp GM$ and G, M, M_A, M_Q all lie on the same circle,

$$AQ = 2M_A M_Q = 2MG$$

56 clubs; 18.5 hours

Problem 25 (18RMM6)

Let the tangents from X intersect Γ at E, F . Let line l meet Γ at T . Let O be the center of Γ .

We invert about Γ . Note that from tangents, we know X, Y, Z goes to the midpoints of EF, TE, TF respectively. We also have Ω goes to an arbitrary circle within Γ such that O is not inside Γ .

We now T is an arbitrary fixed point on Γ , X^* is an arbitrary point on this fixed circle Ω^* , EF is a chord through X^* where X^* is its midpoint meaning $EF \cap X^*O$. The circle $(X^*Y^*Z^*)$ is now the nine-point circle of EFT .

Claim: The nine-point center moves on a circle with positive radius.

Proof: Note that N_9 is the midpoint of OH . We may use complex numbers to find,

$$n_9 = \frac{e + t + f}{2} = \frac{f}{2} + x^*$$

Note that $\frac{f}{2}$ is fixed and x^* moves on a circle with positive radius as desired. $\square$

Note that the radius R of the nine-point circle is always half that of Γ . Therefore, since the nine-point center moves on a circle, say of radius r centered at O_1 , the two circles $(X^*Y^*Z^*)$ is tangent to are the circles centered at O_1 with radius $R \pm r$.

Problem 27 (16HRVTST33)

We invert about (ABC) . Note that $(AOE), (AOF)$ go to lines AF', AE' . M', N' are where the external angle bisectors of $\angle AOE', \angle AOF'$ meet AE', AF' respectively. We show that,

$$\begin{aligned} 2\angle BAC &= \angle ENO + \angle OMF \Leftrightarrow \angle BOC = \angle ENO + \angle OMF \\ &\Leftrightarrow \angle E'OF' = \angle OE'N' + \angle F'OM' \Leftrightarrow E'N' \parallel F'M' \Leftrightarrow \frac{E'A}{AM'} = \frac{N'A}{AF'} \end{aligned}$$

Indeed, we have $\frac{E'A}{AM'} = \frac{E'M'}{AM'} - 1 = \frac{E'O}{AO} - 1 = \frac{AO}{EO} - 1 = \frac{BO}{EO} - 1 = \frac{BE}{EO}$.

Similarly, $\frac{N'A}{AF'} = \left(\frac{N'F'}{AF'} - 1\right)^{-1} = \left(\frac{OF'}{AO} - 1\right)^{-1} = \left(\frac{AO}{OF} - 1\right)^{-1} = \frac{FC}{OF}$.

These two quantities are given to be the same.

66 clubs; 22 hours

Problem 28 (14TWNTST3J3)

Let D, E, F be the contact points. We invert about the incircle. A, B, C go to the midpoints of DE, FD, EF respectively. We know M^* is an arbitrary point on Γ^* now the nine-point circle of DEF . Let the points where the tangents from M met the incircle be P_1, P_2 respectively.

We know that M^* is the midpoint of chord P_1P_2 meaning P_1, P_2 are where the line perpendicular to M^*I through M^* meet (DEF) . We have X_1^*, X_2^* are the midpoints of P_1D, P_2D as they are the intersections of the tangents at D and P_1 or D and P_2 . Where M_A is the midpoint of major arc BC , we know that the tangency-point T of the A -mixtilinear circle is collinear with I, M_A .

Note that,

$$\begin{aligned}\angle BAI + \angle ITB &= \frac{\angle A}{2} + \frac{\angle B + \angle C}{2} \\ &\Leftrightarrow \angle BAI + \angle ITB = 90 \\ &\Leftrightarrow \angle T^*B^*I + \angle IB^*A^* = 90\end{aligned}$$

Therefore, A^*T^* is a diameter of the nine-point circle. If we let H be the orthocenter of DEF , T^* is the midpoint of DH . We show $(X_1^*M^*X_2^*)$ passes through this point, which would suffice.

We perform a homothety at D of scale two. Since $(X_1^*M^*X_2^*)$ was previously also the nine-point circle of DP_1P_2 , it must now be the reflection of (DP_1P_2) over P_1P_2 . It suffices to show that the reflection of H over P_1P_2 lies on (DEF) . By a homothety at H , it suffices to show that the base of the altitude from H to P_1P_2 lies on $(A^*B^*C^*)$. This is clearly true by symmetry about the nine-point center.